IMATINIB

Take with food & large glass of water

Fluid retention

CHEMICAL NAME 4-[(4-methylpiperazin-1-yl)methyl]-N-[4-methyl-3-[(4-pyridin-3-ylpyrimidin-2-yl)amino]phenyl]benzamide (mesilate)

CLASS Tyrosine kinase inhibitor (BCR-ABL, KIT, PDGFR)

TRADE NAMES Glivec · Gleevec · Imatib · Veenat

THERAPEUTIC USES / INDICATIONS

- Chronic myeloid leukaemia (CML) — Philadelphia chromosome-positive.
- Ph+ acute lymphoblastic leukaemia.
- Gastrointestinal stromal tumours (GIST) (KIT-positive).
- Hypereosinophilic syndrome, systemic mastocytosis, dermatofibrosarcoma protuberans.

ADMINISTRATION

- **PO** once daily with a meal and a large glass of water (↓ GI irritation).
- Tablets can be dispersed in water or apple juice.
- Cytotoxic handling (avoid crushing).

DOSAGES

- **CML chronic phase:** 400 mg once daily; accelerated/blast phase 600–800 mg.
- **GIST:** 400 mg daily (800 mg for KIT exon 9 mutations).
- **Children:** 340 mg/m²/day.
- Adjust for cytopenias, hepatotoxicity.

PHARMACOKINETICS

- **Peak:** 2–4 h. Bioavailability ~98%.
- Metabolised by CYP3A4 to active metabolite.
- Excreted in faeces. **t½:** 18 h (metabolite 40 h).

PHARMACODYNAMICS

- Competitively inhibits ATP binding to **BCR-ABL tyrosine kinase** (and KIT, PDGFR) → blocks proliferation signalling → apoptosis of malignant cells.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Fluid retention (periorbital oedema), weight gain.
- Nausea, diarrhoea, muscle cramps.
- Rash, fatigue.

ADVERSE EFFECTS (SERIOUS)

- Myelosuppression (early).
- Hepatotoxicity.
- Severe fluid retention (pleural effusion), heart failure.
- GI bleeding (GIST); hypophosphataemia; growth retardation in children.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- Pregnancy, breastfeeding.
- **Caution:** cardiac disease, hepatic impairment.

KEY INTERACTIONS

- **Strong CYP3A4 inhibitors (ketoconazole, clarithromycin, grapefruit):** ↑ levels.
- **Inducers (rifampicin, St John's wort, phenytoin):** ↓ efficacy.
- **Simvastatin, ciclosporin:** ↑ their levels.
- **Paracetamol (high dose):** hepatotoxicity; **warfarin:** use LMWH.

NURSING MANAGEMENT

- FBC weekly first month, fortnightly second month, then periodically.
- Monitor weight (oedema), LFTs, electrolytes.
- Monitor molecular response (BCR-ABL PCR).
- Adherence counselling (critical for CML outcomes).

PATIENT EDUCATION

- Take with food and a large glass of water every day.
- Weigh yourself regularly; report rapid weight gain, swelling, breathlessness.
- Avoid grapefruit and St John's wort.
- Report fever, bleeding, yellow skin.

KEY POINTS

- BCR-ABL tyrosine kinase inhibitor.
- Fluid retention, cramps, myelosuppression.
- CML and GIST; 400 mg daily with food.
- CYP3A4 interactions; adherence vital.